Ginger Schmidt

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EDUCATION

Harvard-MIT Health Sciences and Technology — Cambridge, MA

Ph.D. Medical Engineering and Medical Physics — May 2027

- MIT Electrical Engineering concentration
- Thesis: "Minimally invasive asynchronous optical coherence elastography for cardiology and ophthalmology"
- MIT Coursework: Optics & Photonics, Nonlinear Optics, Advances in Computer Vision (Neural Networks), Digital Control Systems, Biomedical Signal & Image Processing
- Harvard Medical School Coursework: Cardiovascular Pathophysiology, Respiratory Pathophysiology, Genetics, Immunology, Human Pathology

Harvey Mudd College — Claremont, CA

B.S. Engineering — May 2021

- Coursework: Robotics State Estimation, Robotics Motion Planning, Advanced Systems & Signal Engineering, Dynamics of Elastic Systems, Differential Equations, Linear Algebra, Data Structures (C++), Digital & Analog Electronics

RESEARCH EXPERIENCE

Massachusetts General Hospital and Harvard Medical School — Boston, MA

Graduate student researcher — June 2021 – present

- Developing signal processing methods to extract quantitative tissue dynamics from OCT electromagnetic phase measurements for multi-frequency, 3D mechanical characterization.
- Designing custom miniaturized micromotor-based imaging catheter for ultrafast coronary artery imaging.
- Inverse methods and optimization for solving wave-propagation problems in isotropic and transversely isotropic media to recover tissue mechanical properties from measured displacement fields.
- Advised by Professors Néstor Uribe-Patarroyo and Brett E. Bouma

Lab for Autonomous and Intelligent Robotics — Claremont, CA

Undergraduate student researcher — Sept 2019 – May 2021

- Designed custom centimeter-scale autonomous underwater glider that conducts electrolysis of water to modulate its center of mass, powered by miniature watch cell battery.
- Performed system identification, controls/stability simulations, and validated autonomous depth tracking.
- Advised by Professors Christopher Clark and Matthew Spencer.

FELLOWSHIPS

2026-27	MIT Next-Generation Medical Imaging Graduate Fellowship
2025-26	MIT Health and Life Sciences Collaborative Graduate Fellowship (HEALS)
2022-25	National Science Foundation Graduate Research Fellowship (NSF-GRFP)
2021-22	MIT Presidential Fellowship
2017-21	Harvey S. Mudd Merit Scholarship
2017	National Merit Scholarship

AWARDS & HONORS

2025	Best Presentation (International Tissue Elasticity Conference)
2025	Best Poster Award (SPIE Photonics West, Elastography & Tissue Mechanics)
2024	Best Poster Award (Gordon Research Conference, Optics in Medicine)
2023	Best Poster Award (Biophotonics Summer School)
2022	50 Women in Robotics You Need to Know About (Robohub)
2021	Johnson Excellence in Engineering Award (Harvey Mudd College)
2020	Alford-Gilkeson Clinic Award (Harvey Mudd College)
2016	Math Prize for Girls, Top 300 in North America
2015	American Mathematics Competition (AMC), Distinguished Honor Roll, Top 1%

PUBLICATIONS

1. **G. Schmidt,** R. McAuley, B. E. Bouma, and N. Uribe-Patarroyo, "Multi-frequency asynchronous optical coherence elastography by direct demodulation," Biomedical Optics Express 17, 4947-4967 (2026)
2. **G. Schmidt,** R. McAuley, B. E. Bouma, and N. Uribe-Patarroyo, "Asynchronous optical coherence elastography and directional phase gradient analysis," Journal of Biomedical Optics (2025).
3. M. Singh, J. Situ, C. Long, D. Quevedo-Moreno, **G. Schmidt,** N. Uribe-Patarroyo and E. T. Roche, "Coronary Artery Flow Modeling in an Ex-vivo Biorobotic Heart," 2025 IEEE 8th International Conference on Soft Robotics (2025).
4. **G. Schmidt,** B. E. Bouma, and N. Uribe-Patarroyo, "Asynchronous, semi-reverberant elastography," Optica 11, 1285-1294 (2024).

INVITED TALKS

1. Institute of Biomedical Optics at the University of Lübeck, Germany: "Asynchronous optical coherence elastography with a commercial frame rate OCT system." (2026)
2. Optica Biophotonics Congress in Fort Lauderdale, Florida: "Shear wave elastography through a helically-scanned catheter using asynchronous optical coherence elastography." (2026)
3. SPIE Photonics West in San Francisco, CA: "Towards intraluminal elasticity measurements via an endoscope with asynchronous optical coherence elastography." (2026)
4. MIT Medical Imaging Seminar Series in Cambridge, MA (2026)
5. New England Symposium for Biomedical Optics (NESBO) in Cambridge, MA (2025)
6. Wellman-University of Tokyo Graduate Student Symposium in Tokyo, Japan (2025)
7. Drexel University ECE-101 Guest Lecturer in Philadelphia, PA (2023)

CONTRIBUTED TALKS

1. SPIE Photonics West in San Francsico, CA: "Non-destructive mechanical phenotyping of 3D cartilage tissue models for osteoarthritis with asynchronous optical coherence elastography." (2026)
2. SPIE Photonics West in San Francsico, CA: "Biomechanical assessment of coronary arteries under recapitulated physiological motion with asynchronous shear wave optical coherence elastography." (2026)
3. International Tissue Elasticity Conference in Seattle, WA: "Elasticity measurement of the human cornea in vivo with 3D asynchronous optical coherence elastography." (2025)
4. International Tissue Elasticity Conference in France: "Asynchronous semi-reverberant elastography." (2024)
5. SPIE Photonics West in San Francsico, CA: "Overcoming the spatial and temporal coherence limitations of reverberant elastography in raster-scanned OCT systems." (2023)
6. SPIE Photonics West in San Francsico, CA: "Enabling quantitative shear wave elastography in conventional optical coherence tomography in vivo." (2023)

PATENTS

1. Shishkov, M., Bouma, B. E., Uribe-Patarroyo, N., **Schmidt, G.,** "Micromotor and optical arrangement for fast circular scanning of light beams in small diameter flexible catheters." US Application US20260090704A1, European Application EP4590180A2, Japanese Application JP2025529574A. (Pending)
2. **Schmidt, G.,** Uribe-Patarroyo, N., Bouma, B. E., "Tunable shear wave front generation for biomechanical measurement of biological tissue." World Application WO2025160583A1. (Pending)

INDUSTRY, ENGINEERING, & ROBOTICS

Battlebots 2017 – present
Co-captain of robot Tantrum Mountain View, CA

- Season 6 World Championship Winner of Battlebots on Discovery channel and HBO Max (seasons 3-7).

Pressona 2025 – present
Founding engineer Cambridge, MA

- Our core technology is a closed-loop "autopilot" for arterial blood pressure management in the OR and ICU which combines real-time signal processing with adaptive control to help clinicians keep patients in a safe hemodynamic window while reducing the cognitive burden of constant titration.
- In collaboration with CEO Taylor Baum, based on her MIT Ph.D. thesis work with Professor Emory Brown.

Johnson & Johnson, Robotics and Digital Surgery Santa Clara, CA
Mechanical Engineering & Robotics Intern August 2020 – May 2021

- Characterized magnetic crosstalk interference in neighboring encoders.
- Designed 6-axis motor fixture with torque sensors and encoders for surgical instrument manipulator verification & validation testing. Selected electrical and control system components for the fixture.

Auris Health (acquired by Johnson & Johnson) Redwood City, CA
Mechanical Engineering & Robotics Intern, Advanced Development May 2018 – January 2019

- Designed electro-mechanical packaging solution for a surgical instrument manipulator to fit torque sensors in a constrained volume while improving DFM and DFA.
- Mechanical and electrical assemblies of surgical robotic arms with brushless motors, sensors, slip rings, and harmonic drives.

SUPERVISING & MENTORSHIP

Isabel Sperandio '25	B.S. Mechanical Engineering, MIT.
	Current Ph.D. Student in Mechanical Engineering at MIT, continuing in our lab.
Huda Abdelghani '27	B.S. Mechanical Engineering, MIT.
	Best Poster at the New England Symposium for Biomedical Optics (2025).
Eric Zou '29	B.S. Electrical Engineering, MIT.

PROFESSIONAL SERVICE

2023-	Reviewer for Optics Letters, Biomedical Optics Express, Science Robotics
2025-	MIT Next-Generation Medical Imaging committee member, organizing institute-wide monthly seminars
2025-	Graduate Student Advisory Group: Advisor to the Dean of the MIT School of Engineering
2025-	HST Joint Council, MD Curriculum Committee Member
2025-	Graduate Admissions Interviewer for the Harvard-MIT Health Sciences & Technology Ph.D. program
2021-24	MIT Application Assistance Program: Mentored underrepresented students in grad school applications.
2018-21	Prison Education Project: President and instructor, computer science and physics.
2019-21	Resident Assistant and Dorm Proctor at Harvey Mudd College.

TEACHING

Spring 2021	HMC E206: Robotics State Estimation, TA
Spring 2021	HMC E102: Advanced Signals & System Engineering 2, Head Grader and Tutor
Fall 2020	HMC E205: Robotics State Estimation, TA
Fall 2020	HMC E101: Advanced Signals & System Engineering 1, Head Grader and Tutor
Fall 2020	HMC E79: Introduction to Systems Engineering, TA
Spring 2020	HMC E80: Experimental Engineering, Lab Proctor
Fall 2019	HMC E79: Introduction to Systems Engineering, TA